



EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	% / CGPA	Completion
MTech. (Computer Science & Eng.)	Indian Institute of Technology Madras	8.00	2026
B.Tech. (Computer Science & Eng.)	Indian Institute of Information Technology – Kancheepuram	8.02 / 10	2023
Class XII (CBSE)	Maharishi Vidhya Mandir Chetpet, Chennai	91.8%	2019
Class X (CBSE)	Maharishi Vidhya Mandir Chetpet, Chennai	9.4 / 10	2017

PROFESSIONAL EXPERIENCE

Computer Vision Intern - TIHAAN Dr. Rahul Raman Oct'22 – Jun'23	Project title: Monocular Depth Analysis Controlled GPS Denied AGV Navigation for Seamless Tracking - Proposed a geometry-based ML model (PDM) for object trajectory prediction with monocular camera - Created a real time 3-model pipeline using YOLO v4 and Deep-SORT and our proposed PDM model. - PDM model 34% more precise than the 6 state-of-the-art baselines at the time of publishing my paper. <i>Skills: Computer Vision, Deep Learning</i>
Data Scientist Intern Bajaj FinServ (Jun - Aug'22)	Determined if a potential customer if taking loan can pay it back - Worked on estimating if a prospective Policy Holder is financially able to take the policy (Loan) and pay back, and measure the possible success rate of the ground team on converting the prospects. - Used Chi-square Analysis, PCA and optimal KNN clustering (via elbow method) for this project - Address fraud detection <i>Skills: Un-supervised Machine Learning</i>
TA Dr. Manikandan Narayanan (Jun - Nov'24)	As head TA for Programming and Data Structures Lab (CS2710) As Head TA, ensured smooth lab sessions, double-verified test case generation for coding assignments, addressed grading queries, streamlined communication, coordinated with 14 TAs, and maintained timely deadlines to enhance course efficiency

PUBLICATIONS AND CONFERENCES

TCE.2023.3318050 Journal Publication <i>B.Tech Thesis</i>	"Perspective Distortion Model for Pedestrian Trajectory Prediction for Consumer Applications," in <i>IEEE Transactions on Consumer Electronics</i> , S. Gundreddy, <u>R. Ramkumar</u> , R. Raman, K. Muhammad and S. Bakshi, • This is a Govt. funded project from TIHAAN IIT-H.
<i>Draft**</i>	As a conclusion to TIHAAN - sponsored project that we implement a scalable physical solution for tracking and navigation on a AGV by combining the PDM Model and a Depth Estimation model that was developed earlier

Projects

HEART DISEASE PREDICTION <i>Course Project</i>	Predict the possibility of heart disease in a person based on many life-style characteristics - Worked on the Data pre-processing; Dealt with Correlation, Outliers and Imbalanced Data by resampling - Used Random Forest Model to achieve 98% Cross Validation Score and F1 Score of 0.95
Credit Card Fraud Detection	Detect possible credit card frauds in an extremely skewed and imbalanced data set - Formulated a model to detect Credit card frauds, with an accuracy of 99.89 % in an imbalanced dataset where only 0.17% of the 2.5 lakh transaction were fraudulent with Logistic Regression - Mitigated skewness by bringing them into normal distribution by power transformation - The model achieved F1 score of 0.65 and ROC score of 0.99
TOMSULO ARCH. SINGLE CORE PROCESSOR <i>Course Project</i>	Build a Tomsula Arch. with Verilog and Python Building arithmetic modules like 64-bit Floating Point Adder and Multiplier. Regular Adders Multipliers, Arithmetic Logic Unit and Register File in Verilog HDL and using the above mentioned as base and building a Tomsula Arch. Single Core Processor in Python

Courses and Certifications

Cloud Computing	Cloud Computing NPTEL	Azure Fundamentals* AZ 900	Azure Administrator* Az-104		
AI and ML	Deep Learning IIT Ropar	Visualizing Data with Python (IMB)	Analytics and Systems of Big Data (ASBD)	Computer Vision	
Quantitative Analysis	Managerial Economics - NPTEL		Linear Algebra and Random Processes (LARP)		
FSD (GfG)	Full Stack Development with React & Node JS				
Skills	Language- Python, SQL, C++, HTML5, CSS, VHDL, MIPS32 (Assembly Language) Skills – SDE (C++), ML, Computer Vision (CV)				

EXTRA-CURRICULARS

Bookworm	Love to read books, especially thriller mystery and Fantasy novels
Freshers Night	Was a member of Sponsorship Team for freshers-night, secured Red bull Sponsorship

* Ongoing Courses in current semester

** The Research Paper is in Draft Stage, it will be published in the coming months